01 M2 DECISION RECORD

Rendered evidence closes M2

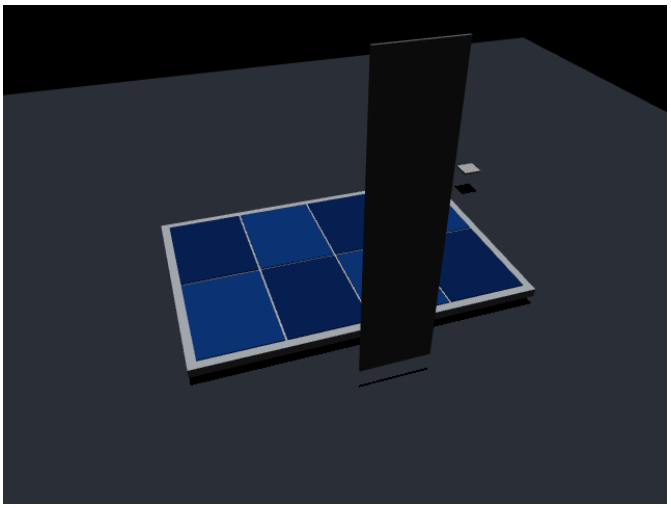
Observation/action integrity, finite visibility fixtures and delayed RGB delivery.

PASS_M2

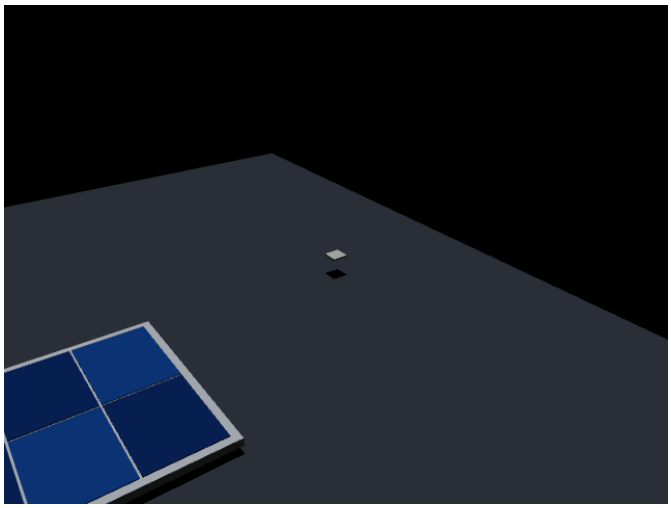
Independent desktop gate reviewed the retained run. Stop before M3.

519	33	29	27
tests passed	render cases	original PNGs	JSON records

The full development suite passed with zero failures, errors or skips. It executed 486 non-render and 33 real-render cases. The exact frozen 519-case inventory matched, and all 84 recorded effective source/input hashes remained unchanged. [S1, S2]



Narrow screen / lateral view



Reframed reference patch

What the image count means

24 direct snapshots compare production and calibration geometry at actual settled poses. Five sensor images comprise three actual exposure-midpoint captures plus two evaluator-only delivery-pose comparisons. These are bounded deterministic fixtures, not new complete policy pilots. [S4, S5]

Execution: 18 October 2026, 04:36:57 UTC | source revision 34eee53dc38a | desktop macOS / MuJoCo 3.3.7

02 SCIENTIFIC SCHEMATIC / NOT RGB

Geometry explains the visibility cases

Coordinates and camera centres from retained configuration and pose records. [S3, S4, S8]

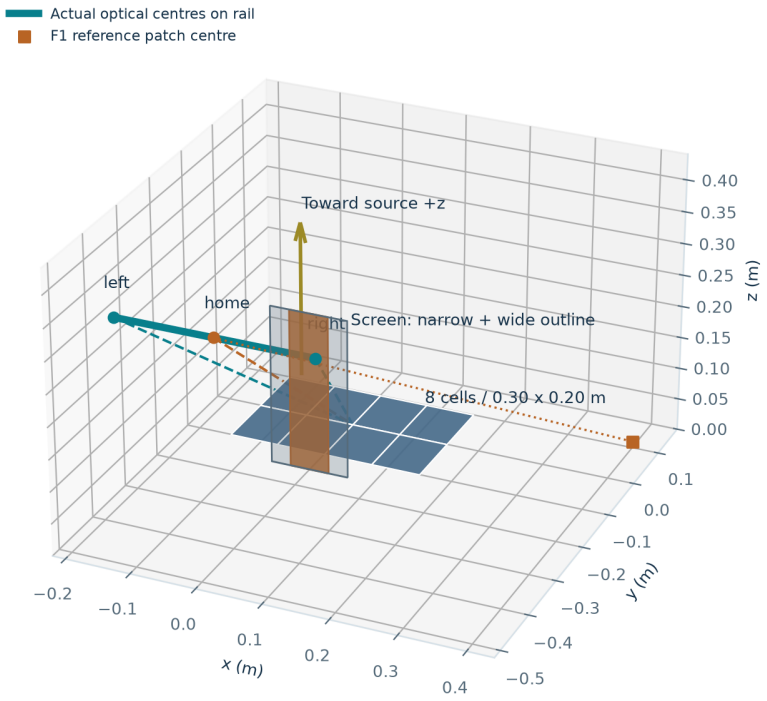


Figure 1. Actual optical centres lie on a 0.30 m rail at y=-0.45 m, z=0.35 m. Dashed rays connect those centres to the panel centre; the F1 reference ray uses the displaced patch. This Matplotlib schematic is explanatory geometry, not a camera image.

Fixture	Declared geometry	Interpretation
F1 reframe	No screen; patch x=0.40 m	Rotate at fixed centre; patch enters FOV.
F2 narrow	Screen half-width 0.03 m	Endpoint rays clear the panel centre.
F2 rail locked	Same narrow screen; x=0 only	Rotation retains the blocked target ray.
F2 wide	Screen half-width 0.06 m	Target remains blocked over the full rail.

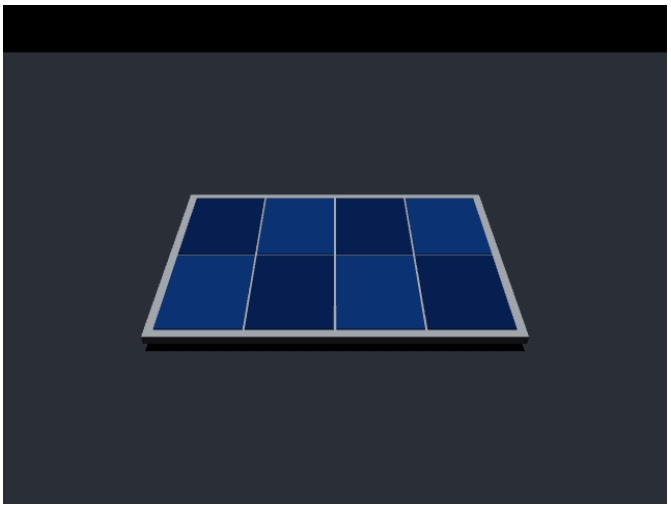
Screen centre (0,-0.15,0.125) m; half-depth 0.001 m; half-height 0.125 m. At the screen faces, endpoint centre-ray |x| spans 0.049667-0.050333 m: clear of narrow, inside wide. Toward-source direction is +z; all 8 cells remain at 900 W/m2. These screens occlude camera views without adding source shading. [S7, S8]

03 ACTUAL MUJOCO RGB / DIRECT POSE SNAPSHOTS

F1: rotation recovers framing

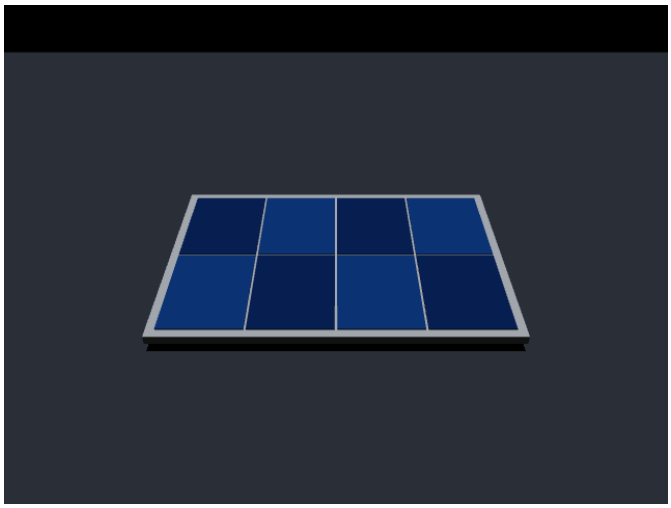
Four original images; home and reframed poses share the same optical centre.

PRODUCTION / home



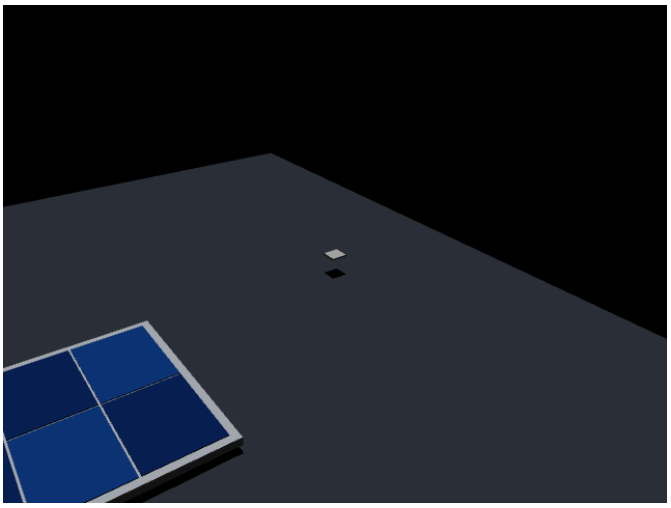
t=8 s; q=(+0.000 m, +0.000, +0.000 rad)

CALIBRATION / home



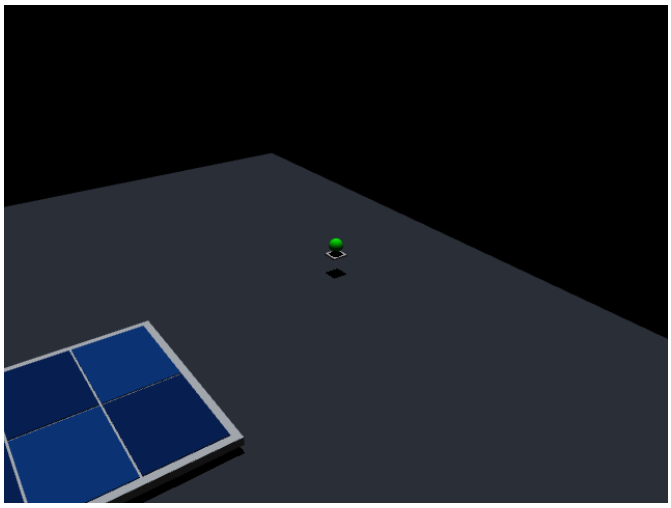
t=8 s; q=(+0.000 m, +0.000, +0.000 rad)

PRODUCTION / reframed



t=16 s; q=(-0.000 m, -0.629, +0.197 rad)

CALIBRATION / reframed



t=16 s; q=(-0.000 m, -0.629, +0.197 rad)

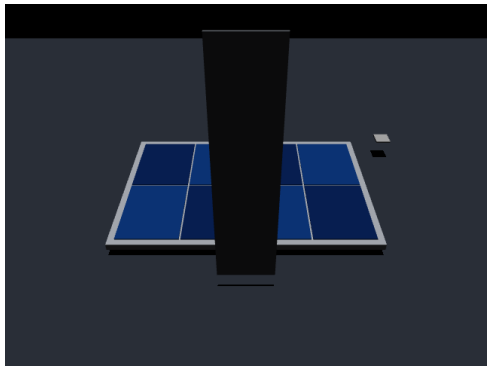
The home view misses the reference patch at (0.40, 0.10, 0.01) m. Bounded yaw/pitch brings it into frame at the same optical centre. The bright green calibration locator is separate from production RGB. [S4, S8]

04 ACTUAL MUJOCO RGB / DIRECT POSE SNAPSHOTS

F2: translation clears a narrow screen

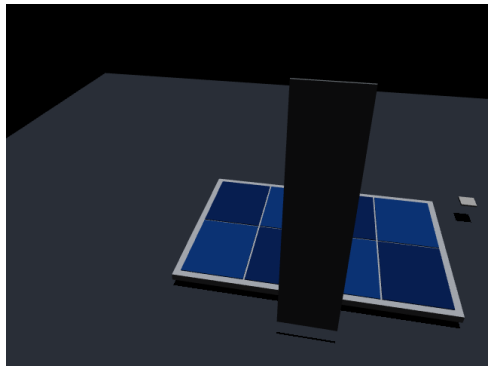
Eight original images. Green calibration locator appears at both lateral endpoints; home and rotation remain blocked.

PRODUCTION / home



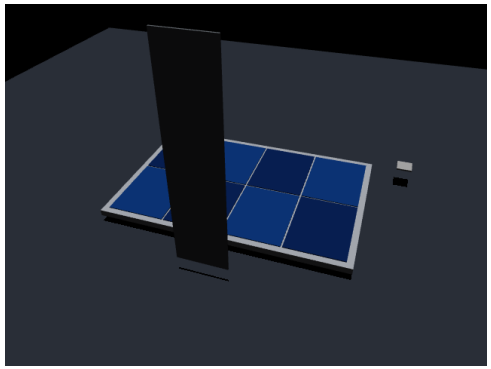
t=8 s; x=+0.000 m
yaw=+0.000; pitch=+0.000 rad

PRODUCTION / rotation only



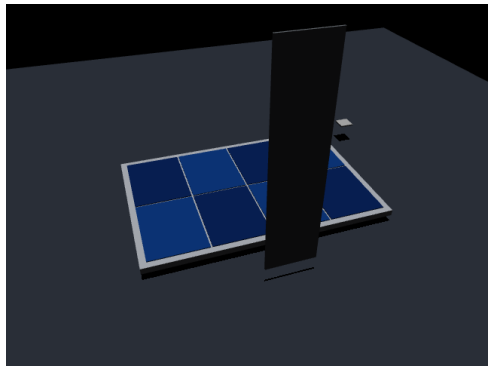
t=16 s; x=+0.000 m
yaw=+0.200; pitch=+0.100 rad

PRODUCTION / right



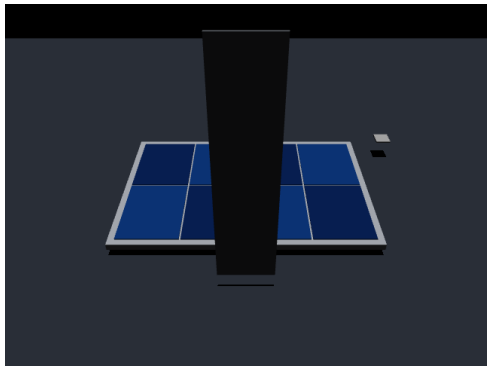
t=24 s; x=+0.150 m
yaw=+0.322; pitch=+0.025 rad

PRODUCTION / left



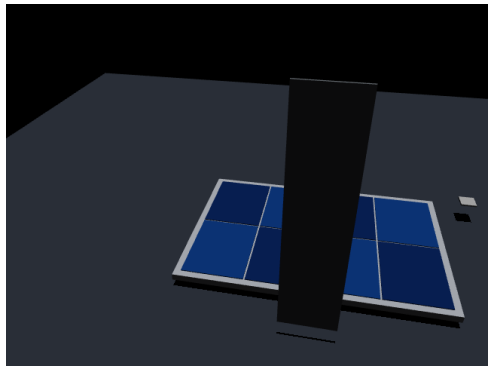
t=32 s; x=-0.150 m
yaw=-0.322; pitch=+0.025 rad

CALIBRATION / home



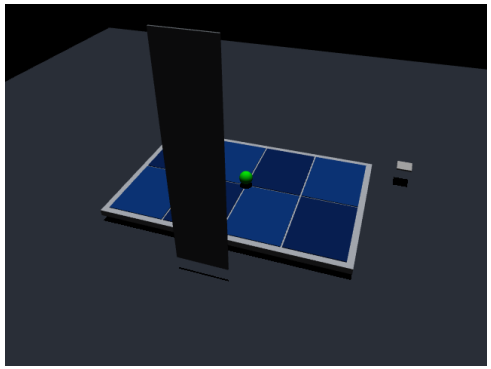
t=8 s; x=+0.000 m
yaw=+0.000; pitch=+0.000 rad

CALIBRATION / rotation only



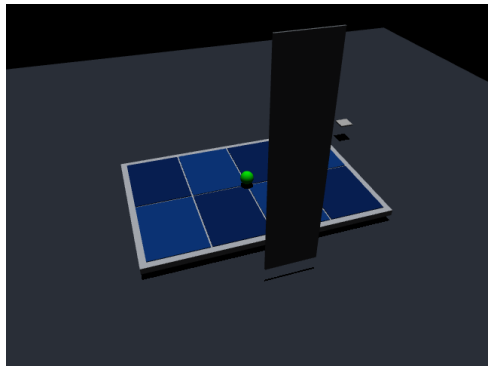
t=16 s; x=+0.000 m
yaw=+0.200; pitch=+0.100 rad

CALIBRATION / right



t=24 s; x=+0.150 m
yaw=+0.322; pitch=+0.025 rad

CALIBRATION / left



t=32 s; x=-0.150 m
yaw=-0.322; pitch=+0.025 rad

Production contains no green fiducial. Calibration adds a locator solely to test projection/visibility; identical blocked views can therefore share the same PNG hash. [S4, S8]

05 ACTUAL MUJOCO RGB / DIRECT POSE SNAPSHOTS

F2: the wide screen blocks the full rail

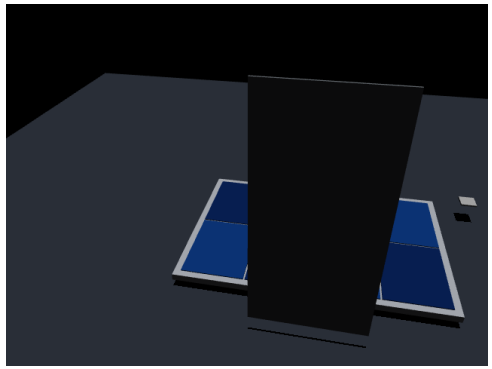
Eight original images. The same finite rail endpoints cannot clear a screen of 0.06 m half-width.

PRODUCTION / home



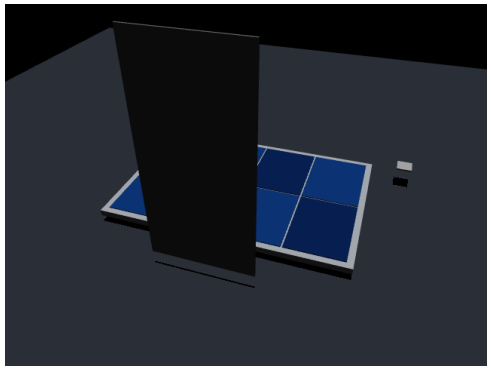
t=8 s; x=+0.000 m
yaw=+0.000; pitch=+0.000 rad

PRODUCTION / rotation only



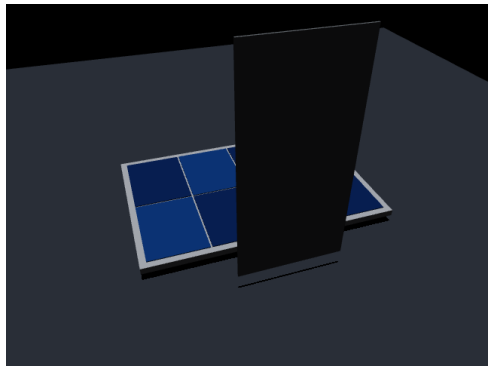
t=16 s; x=+0.000 m
yaw=+0.200; pitch=+0.100 rad

PRODUCTION / right



t=24 s; x=+0.150 m
yaw=+0.322; pitch=+0.025 rad

PRODUCTION / left



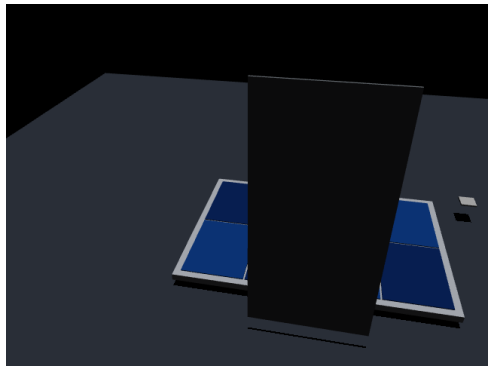
t=32 s; x=-0.150 m
yaw=-0.322; pitch=+0.025 rad

CALIBRATION / home



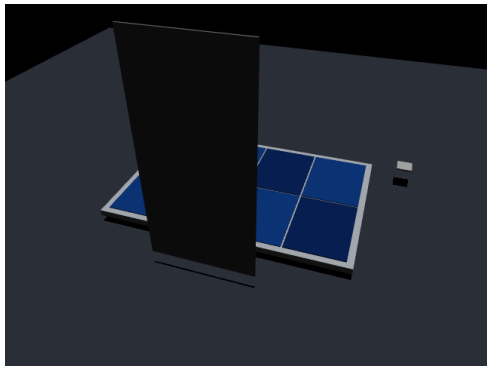
t=8 s; x=+0.000 m
yaw=+0.000; pitch=+0.000 rad

CALIBRATION / rotation only



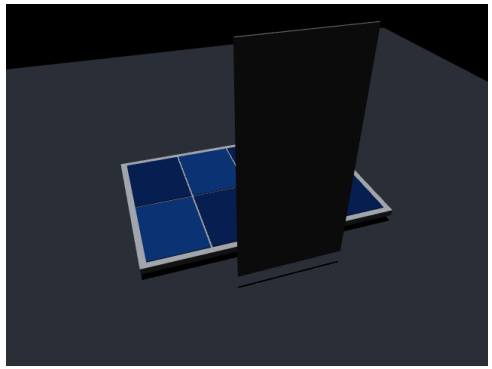
t=16 s; x=+0.000 m
yaw=+0.200; pitch=+0.100 rad

CALIBRATION / right



t=24 s; x=+0.150 m
yaw=+0.322; pitch=+0.025 rad

CALIBRATION / left



t=32 s; x=-0.150 m
yaw=-0.322; pitch=+0.025 rad

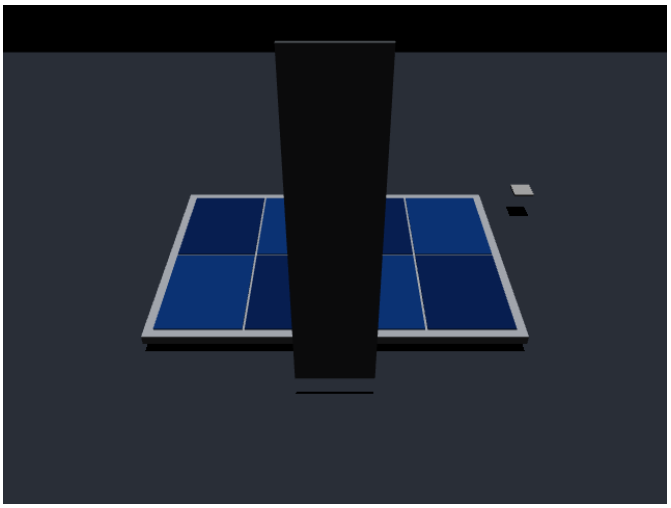
Production contains no green fiducial. Calibration adds a locator solely to test projection/visibility; identical blocked views can therefore share the same PNG hash. [S4, S8]

06 ACTUAL MUJOCO RGB / DIRECT POSE SNAPSHOTS

F2: the rail limit removes translation

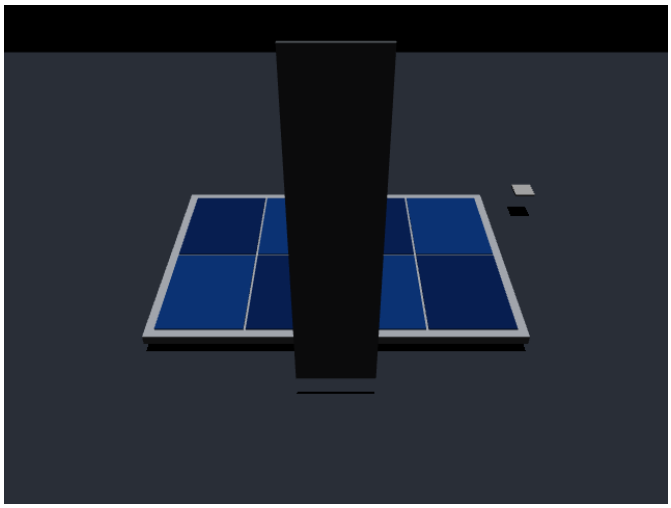
Four original images; production and calibration are separate runs at matching actual poses.

PRODUCTION / home



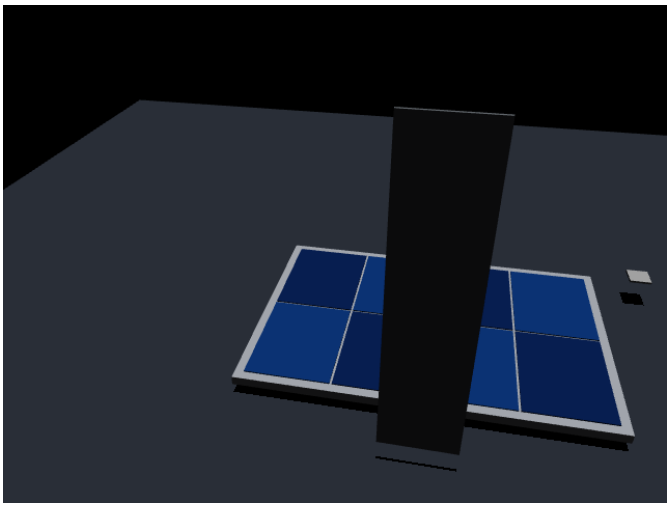
t=8 s; q=(+0.000 m, +0.000, +0.000 rad)

CALIBRATION / home



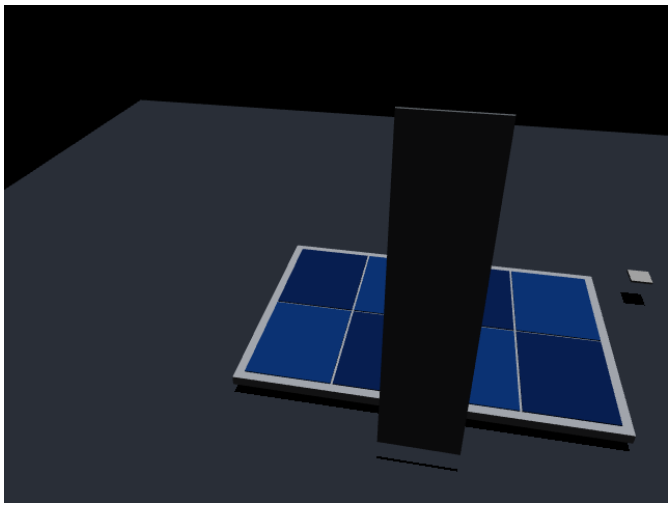
t=8 s; q=(+0.000 m, +0.000, +0.000 rad)

PRODUCTION / rotation only



t=16 s; q=(+0.000 m, +0.200, +0.100 rad)

CALIBRATION / rotation only



t=16 s; q=(+0.000 m, +0.200, +0.100 rad)

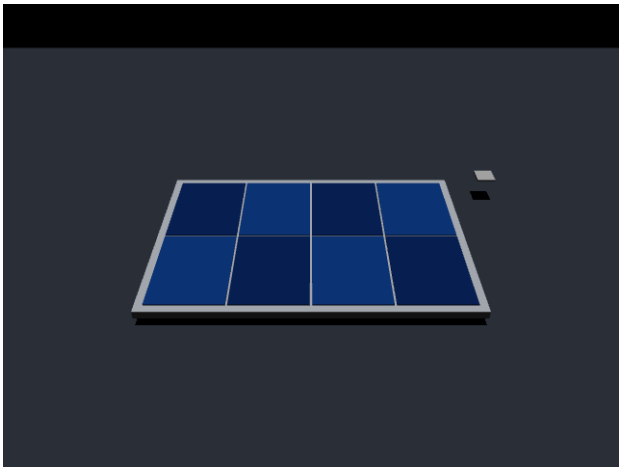
The rail-locked fixture advertises slide limits (0, 0). Pan/tilt can reframe the screen, but cannot change the ray from this optical centre to the same panel point. No lateral endpoint view is available. [S4, S8]

07 ACTUAL SENSOR RENDERS / FIVE RETAINED IMAGES

Historical pixels survive late delivery

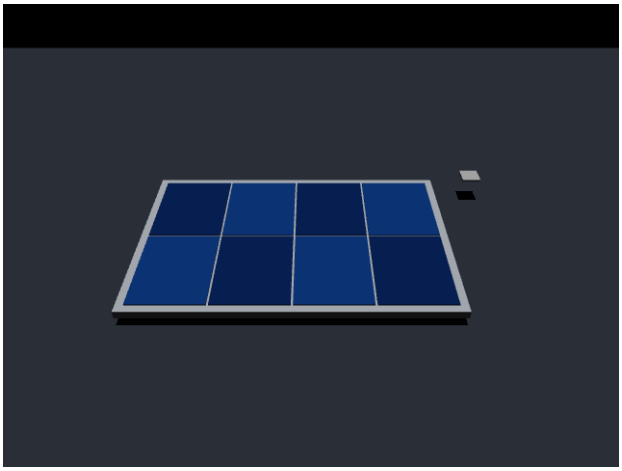
Exposure-midpoint capture and evaluator-only delivery-pose comparison are different evidence roles. [S5]

LATE VALID / ACTUAL MIDPOINT CAPTURE



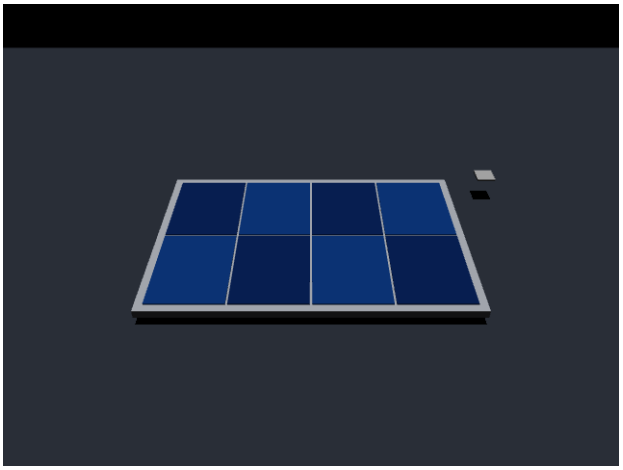
t=0.226 s; slide=0.000 mm

LATE VALID / EVALUATOR-ONLY DELIVERY POSE



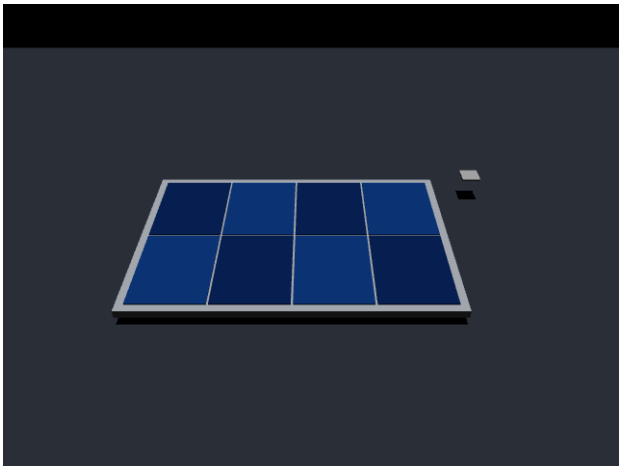
t=1.132 s; slide=16.956 mm

LATE STALE / ACTUAL MIDPOINT CAPTURE

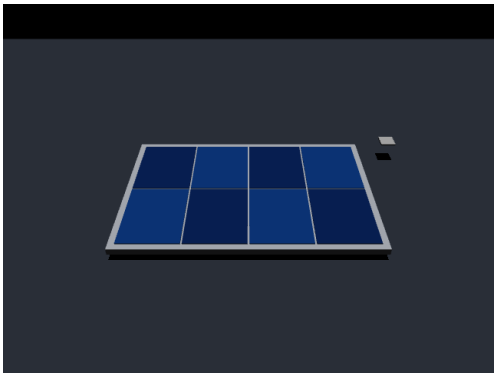


t=0.226 s; slide=0.000 mm

LATE STALE / EVALUATOR-ONLY DELIVERY POSE



t=1.132 s; slide=16.956 mm



DROPPED / ACTUAL MIDPOINT CAPTURE

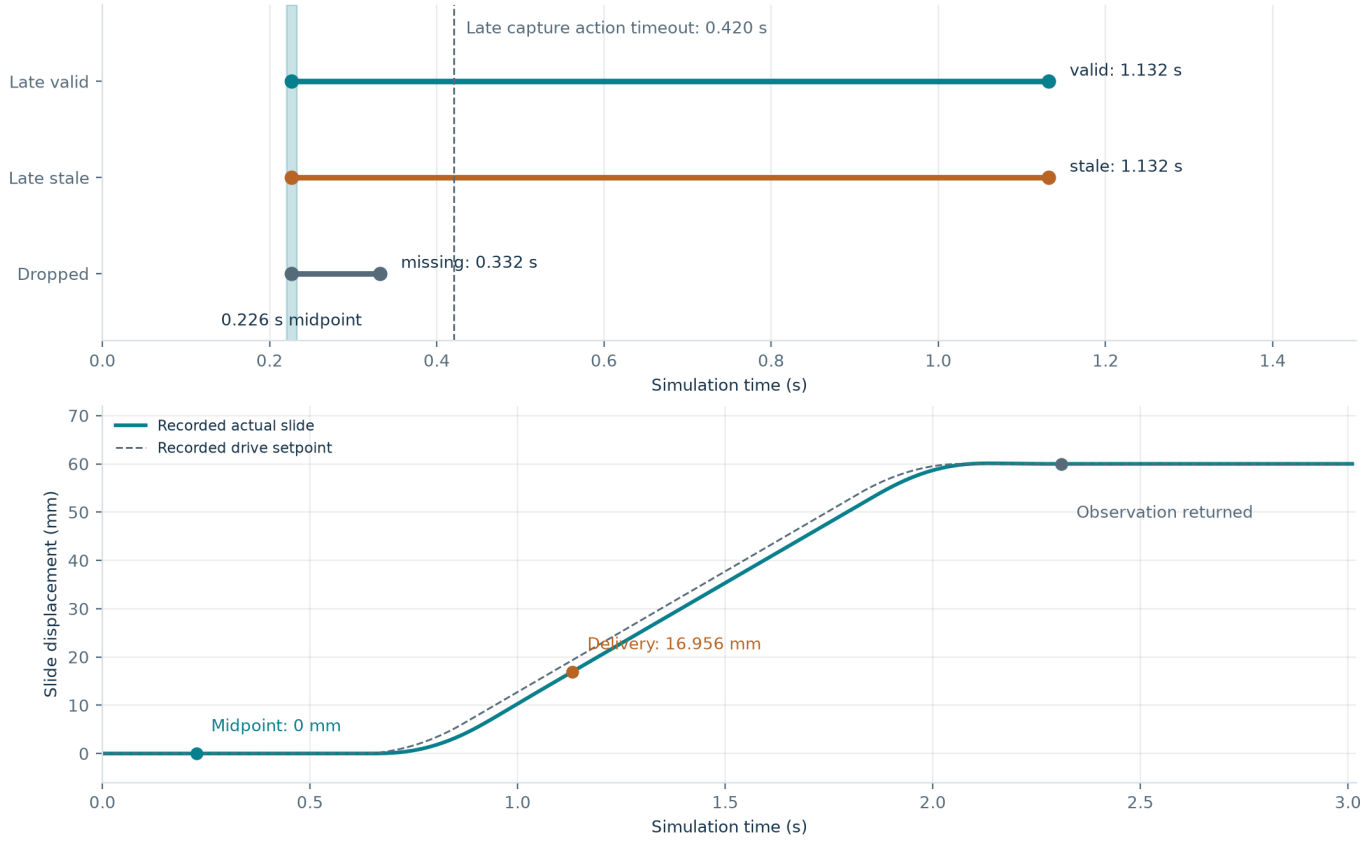
The dropped capture was rendered at 0.226 s and retained privately. The public record delivered at 0.332 s is missing with a null PNG payload. No delivery-pose diagnostic was created for this case.

For late valid and late stale, delivered PNG hashes equal the original midpoint PNG, and differ from the delivery-pose diagnostic. Identical fixture images across cases are expected. Saved timelines store hashes; the executed tests decode public payloads and check pixel equality.

08 TIMING AND IMMUTABLE CAPTURE

Delivery does not recapture the scene

Charts are calculated from the saved timeline JSON and actual dynamics; no simulation was rerun. [S5]



Case	Capture / exposure end	Transport delivery	Returned observation
Late valid	0.226 / 0.232 s	1.132 s; valid	2.310 s
Late stale	0.226 / 0.232 s	1.132 s; stale	2.310 s
Dropped	0.226 / 0.232 s	0.332 s; missing/null	0.332 s

The late cases time out at 0.420 s, then move to x=0.060 m without requesting another RGB capture. Delivery occurs during motion at x=0.016956 m; the policy receives the historical packet in the completed move observation at 2.310 s. A later wait creates no RGB frame.

Each attempted capture consumes one frame: remaining RGB budget is 19/20 in all later observations, including missing/stale cases. Exposure remains 12 ms with a midpoint approximation. Staleness is age at transport delivery, not age at the later observation. [S5, S9]

09 EXECUTION AND CONTRACT INTEGRITY

The desktop result completes the core gate

The full suite count is one execution; overlapping reviewer selections are not added. [S1, S2, S6, S7]

Evidence	Result	Scope
Full desktop regression	519 passed; 33 render; 0 skipped; stdout 25.15 s	Exact inventory; JUnit elapsed 25.092 s
Desktop wrapper	Schema export --check + verification; exit 0	26.339 s total; shared bounded deadline
Independent core review	486 passed; 33 deselected; 19.08 s	Core ready; previous desktop block now addressed
Independent desktop review	PASS_M2	Retained evidence and images reviewed
Short core replay fixture	729 truth rows; exact numeric replay	1.456 s fixture; not a complete pilot

Implemented observation and action contracts

Capture-indexed noise uses versioned seed/channel/tick/purpose inputs. Numeric and encoder noise are separate from transport decisions; RGB pixels remain noise-free. Late, stale and missing records preserve their capture identity. Fresh valid measured feedback is required for confirmation. [S7, S9]

Finite drive-command inhibition models movement stalls. Private load commands can drop or delay; a nominal acknowledgement does not prove application. Bounded restoration/disconnection, cancellation, narrowed advertised limits and terminal no-resume behavior are covered by the core/full regression evidence. [S7, S9]

No additional complete B0/B1 policy pilot was run. Report production used retained PNGs, JSON, scientific plotting and PDF rasterization. The 0.05 m/s slide cap applies to the drive setpoint ramp: recorded finite-servo velocity peaks at 0.051243 m/s. Trusted-policy/local is not hostile-code OS isolation.

10 PROVENANCE, LIMITS AND HANDOFF

A bounded PASS_M2, then stop

Immutable source snapshots precede this PDF; rendering and visual QA are recorded separately.

Source key

Key	Retained source
S1	Desktop verification-report.json and batch-report.json
S2	tests.xml, command logs and frozen m2_test_inventory.json
S3	Expanded private config.json; configs/dev.toml and pilot.toml
S4	24 view PNG/JSON pairs in the retained render/ directory
S5	3 sensor timelines and 5 PNGs in render/sensor_delivery/
S6	M2_DESKTOP_GATE_REVIEW_20261018.txt (fresh desktop gate)
S7	M2_CORE_REVIEW_20261017.txt (prior core review)
S8	m2_fixtures.py, scene.py, optics.py and visibility test source
S9	ADR 0010/0011, M2-core-and-desktop.md; sensor delivery test

Identity and reproducibility

Source revision: **34eee53dc38a4575296d1bf6bfc24501f4e2c5da**
View/verification seed: **20261016**; sensor fixture seed: **61**.
Expanded verification config SHA256: c555351bd7e48d533af87d024d2bdc01813f966f0f124e261c009bcfc717b7ad
Each view JSON also records its fixture config and MJCF hashes; each sensor timeline records its own fixture config hash.

Immutable provenance snapshot SHA256:
6e03a8cedbb7b17076612338596f28954db55bb9f74cd00e9bf9b283bd85deef
runs/m2-desktop-report-20261018/provenance.json
Rebuild: .tools/pdf-report/bin/python scripts/build_m2_evidence_report.py
Use new --work-dir and --output paths; existing outputs are refused.

Retained limits

Extremely large finite private timing values can exceed integer/runtime ranges instead of receiving a clean rejection; this nonblocking core-review limitation remains. Bounded fixtures do not establish general adversarial numeric robustness. This is a deterministic synthetic surrogate: physical calibration, source shading, causal performance, scenario corpus and M3 refinement are outside this acceptance. Stop before M3 for Wei's decision. [S6, S7, S9]